

Episode B2 | *Supply*

A simple way to organize costs.

[ED: Units pass for class: B2 now illustrates one market day, with quantity in kilograms and price in dollars per kilogram. Each narrow animation bar is 1 kg. Part A is unchanged; when reconciling it, use metric tonnes (1 tonne = 1,000 kg). This is a smaller-scale example, not a conversion of A's annual quantities.]

Last time we introduced preferences as a first step to resolving where on the PPF to live, and we used the buyer's Marginal Benefit to decide whether another purchase was worth making. The extra value above what they paid was their Consumer Surplus.

Every exchange has two sides: a buyer and a seller. Just like we asked Amanda-Grace how much she values chocolate, we can just ask the seller how much they would be willing to produce at a range of prices.

Remember Molly? She can sell the spinach she chooses to grow at the price in the market. We'll measure her harvest in tons per year, like we did in Part A.

Let's say Molly is happy to grow 20 kilograms of spinach at a price of \$4 per kilogram. This is Molly's quantity supplied at \$4, which we can write as $Q_s(4) = 20$.

This point represents her ***Individual Quantity Supplied***: the quantity a seller is willing and able to sell at a given price.

Show the definition of quantity supplied.

Like with demand, let's plot this on the XY plane with the price of \$4 on the vertical and 20 kilograms of spinach on the horizontal.

Show the point on the XY plane.

You'll notice that while this graph looks a lot like our PPF, it's doing a very different thing. Instead of plotting spinach against carrots, we're plotting spinach against price.

Why doesn't Molly produce one more? She could definitely sell more if she wished.

Because selling more is only worth doing if the extra spinach brings in enough to cover what it costs her. At this price, she stops at 20 kilograms. Producing a little more would cost more than she would receive for it.

What happens when the price of spinach goes up? More of that extra production becomes worth doing. Molly can devote more land and labor to spinach, giving up their other uses. Let's say she would like to produce 40 kilograms of spinach at a price of \$6 per kilogram. We can plot this on the PQ graph too.

Show more of her farm devoted to spinach, and the point (40, 6) on the same price-quantity graph.

As the price of spinach has gone up, Molly has increased her quantity supplied of spinach since it has become relatively more valuable. What would happen if the price drops, say to \$2? The quantity of spinach Molly would supply would go down because spinach is relatively less valuable than other uses of her land and labor. In this example, at \$2 she would produce none.

We can keep asking this question at different prices, just like we did with Amanda-Grace. Each answer gives us another point.

When we represent this relationship with a curve we call it the ***Individual Supply Curve***. This allows us to answer the question of how much Molly would sell at any price.

Show the definition of Individual Supply Curve.

This positive relationship between prices and quantity is called the ***Law of Supply***.

Law of Supply: *The quantity supplied of a good rises with its price.*

Like demand, supply does not have to be a straight line. But we'll use a linear relationship to keep the math simple. Molly's supply curve in this example is $P = 2 + \frac{Q_s}{10}$.

Connect the three points and write $P = 2 + \frac{Q_s}{10}$. Label the vertical axis dollars per kilogram and the horizontal axis kilograms. Use the nonnegative-quantity part of the line; below \$2, quantity supplied is zero.

Having an equation is handy. We can ask about lots of prices. How much spinach would Molly produce at \$5 per kilogram? Start at \$5 on the vertical axis, move across to her supply curve, and then down to the quantity axis. We find 30 kilograms.

Trace price $\rightarrow$ supply curve $\rightarrow$ quantity. Then show $5 = 2 + \frac{Q_s}{10}$, followed by $Q_s = 10(5 - 2) = 30$.

We can do this at any price.

Repeat at \$4.50: $4.50 = 2 + \frac{Q_s}{10}$ $\{++4.50 = 2 + \frac{Q_s}{10}++\}$ gives us $Q_s = 2.5$ $\{++Q_s = 2.5++\}$. Keep the line and the new quantity visible.

We can also ask the question the other way. Suppose Molly is already producing 10 kilograms. What would it cost her to increase production a little? Start at 10 on the quantity axis and read up to the supply curve. Its height is \$3 per kilogram. That tells us the cost per kilogram of adding a little more spinach at that point.

Highlight $Q = 10$ and trace quantity $\rightarrow$ supply curve $\rightarrow$ price. Show $P = 2 + \frac{10}{10} = 3$. Mark a small increase in quantity beside the point, rather than coloring the whole next kilogram as though it cost \$3.

This is what we call **Marginal Cost**: the cost of producing one additional unit.

Marginal still means “one more.” Because spinach is divisible, we can make that addition small. The curve tells us how costly it is to add more at the quantity Molly has already reached. It does not tell us the total cost of everything she has grown, or the average cost per kilogram.

Show the definition of Marginal Cost, keeping the point at $Q = 10$ on the graph.

For a whole kilogram, the added cost is the cost of going from one quantity to the next, like from zero to one kilogram. That is what we’ll show in a one-kilogram bar. The height of the smooth curve at a single point tells us the cost per kilogram of the next small addition. The cost of the whole kilogram comes from adding up those little costs across the bar.

Now lets go back to our original price of \$4. At 10 kilograms, her marginal cost is \$3 per kilogram. Is producing a little more worthwhile? Yes: the next small addition brings in more than it costs. At ~~1-ten~~ $\{++10\text{ kilograms}++\}$, the additional spinach she produces costs her less than the \$4 $\{++\text{ per kilogram}++\}$ it brings in. At ~~2-tens~~ $\{++20\text{ kilograms}++\}$, the marginal cost exactly equals \$4 per ~~ten~~ $\{++\text{kilogram}++\}$. Beyond ~~2-tens~~ $\{++20\text{ kilograms}++\}$, it costs more than it brings in. That’s why she stops where marginal cost meets the price.

Hold price at \$4. Start at $Q = 1 \{++10++\}$. Sweep along supply toward $Q = 2 \{++20++\}$, showing MC below price, equal to price, then above price. Return to $Q = 2 \{++20++\}$. Read $MC = \$4$ from the graph at $Q = 2 \{++20++\}$; no second substitution.

So not only does the supply curve tell us Molly's quantity supplied, it also tells us Molly's marginal cost. This is the same idea as with the Marginal Benefit in the Demand curve.

This way of seeing costs still uses opportunity cost behind the scenes. Growing more spinach takes land and labor away from other uses. We're now measuring the value of what Molly gives up in money instead of carrots. That includes her own time, even when she doesn't write herself a paycheck.

This tells us that Molly's supply curve *is* her **Marginal Cost Curve**: the cost of adding more spinach at each quantity.

For this exercise, we'll allow fractional pasties, just as we allow fractions of a kilogram of spinach.

Pause for Exercise B2 / Q1: quantity supplied at a price, then marginal cost at a quantity. Use the pumpkin-pasty curve on the exercise sheet.

Producer Surplus

Molly makes some spinach and sells it into a market at some price. How nice is this setup for her? Let's zoom in on the first kilogram of spinach and add things up.

Let's offer Molly \$5 per kilogram. We already found that she would produce 30 kilograms at this price. But let's start at the beginning. We'll use a vertical bar for the amount of spinach between zero and one kilogram, just like the bars we used for Amanda-Grace's chocolate.

Keep $P = 2 + \frac{Q_s}{10}$ and the \$5/kg price line. Focus on the first 1 kg bar, $Q = 0$ to 1. Widen it while the surrounding graph fades; retain the vertical price scale. Show a small "1 kg" width cue: this is magnification, not more spinach. The flat cost line is at \$2.05/kg, the curve's midpoint height for this interval.

She's able to sell this first tiny bit of spinach for \$5 per kilogram. The bar is one kilogram wide, so selling it brings in \$5 of revenue. The cost line divides the bar into two pieces. Below the line is the cost of producing that spinach. Above the line, up to the price, is what Molly receives above that cost.

The cost of producing the first kilogram is \$2.05, the area of the lower piece. She receives \$5, so she gets an extra \$2.95 from this exchange. This extra value to Molly is what we call **Producer Surplus**, which is essentially the extra niceness from the exchange.

Color cost below the cost line and PS above the cost line, keeping the full revenue rectangle visible. Keep the animation's qualitative Revenue, Cost, and PS labels; the spoken numbers need no extra on-screen calculation. Show the definition of Producer Surplus.

Keep the existing price comparison on this one-kilogram bar: $\$5 \rightarrow \$3 \rightarrow \$1.50 \rightarrow \$4 \rightarrow \$5$. Cost stays fixed; revenue and surplus change. Below cost, the bar stays grey and the exchange areas disappear. Return the bar to its original width as the graph fades back in.

The height at one point tells us the cost per kilogram of the next small addition there. The area below the curve tells us the cost of producing the whole amount in the bar. We're showing that cost with a flat line inside each one-kilogram bar.

This is for one very specific kilogram of spinach. To measure all of Molly's producer surplus, we need to look at all the spinach produced and sold at this price.

So what about the second kilogram? We ask the same question: is the next bit of spinach worth producing? Through this second interval, marginal cost is still below the price. The second and third slices both add producer surplus.

Preserve the current one-bar-at-a-time focus sequence through the first ten kilograms, then the quicker cost-and-PS reveals through $Q = 30$. Keep

completed bars visible on each return to the graph, with flat cost boundaries. No additional dollar labels or running tally; calculate the combined area below.

At 30 kilograms, marginal cost has reached the price. Producing any more would cost more than it brings in, so Molly stops. We've found both her quantity supplied and how much extra value she gets from selling it.

Highlight (30, 5). Keep the last kilogram's positive surplus visible: the boundary point has $MC = P$, but the whole interval does not have zero surplus.

Retain the animation's graph-to-math step: $5 = 2 + \frac{Q_s}{10}$, then $Q_s = 10(5 - 2) = 30$. Carry 30 back to the quantity axis before measuring the triangle.

Together, the producer surplus bars trace the shape of a triangle.

We can find all of that surplus at once by finding the area of the triangle. For buyers, we looked below demand and above price. For sellers, we look above supply and below price, over the quantity sold.

Highlight the outside boundary of the PS triangle. Keep the vertical bar divisions and their flat cost boundaries visible.

The height is the price minus the vertical intercept of supply. Here that's \$5 minus \$2, or \$3 per kilogram. The base is the quantity supplied, 30 kilograms.

Mark height $\$5 - \2 and base 30. Move the values into $PS = \frac{1}{2}hb = \frac{1}{2}(5 - 2)(30) = 45$.

Molly's producer surplus is \$45. The triangle and the bars measure the same extra value.

Pause for Exercise B2 / Q2: find and label producer surplus on the pumpkin-pasty supply curve at 10 galleons.

Market Supply

Molly isn't the only farmer growing spinach so we need to look at other farmers' individual supply curves as well. You remember Andrew, right? The farmer next door. He also has an individual supply curve.

Let's start at \$4 per kilogram. Molly would supply 20 kilograms and Andrew would supply 10. How much spinach would the two of them supply together? Thirty kilograms. We add the quantities at the same price.

Show Molly's $P = 2 + \frac{Q_s}{10}$ and Andrew's $P = 2 + \frac{Q_s}{5}$ on separate graphs. Hold one horizontal price line at \$4. Move their quantity readouts, 20 and 10, into the market graph to make (30, 4). Pause for the class to add them before revealing the total.

At \$6, Molly would supply 40 kilograms and Andrew would supply 20, giving us 60 kilograms altogether. The higher price makes more production worthwhile for each of them.

Repeat the horizontal addition at \$6: $40 + 20 = 60$. Keep both individual quantities attached to the same price.

When we add together all the farmers individual supply curves, we call this the **Aggregate Market Supply Curve** or sometimes simply **Supply**. This shows how we horizontally sum the quantities of two individual supply curves, but we could do it for any number of sellers.

Molly and Andrew are only two of the sellers. We can keep adding the other farmers' quantities at each price. Just as we did for market demand, we'll represent the full market with a simple line.

Every seller is responding to the same price, and the market curve collects their decisions.

Just like Molly's Producer Surplus, we can find the producer surplus for the entire market as the area above supply and below price, over the quantity sold.

Briefly shade the surplus area on the existing combined-sellers graph at \$4. Keep any bar divisions visible. No new numerical example, running tally, or area calculation.

Closing

Prices offer one coordination device to facilitate exchanges between buyers and sellers. But to answer the big questions we have remaining, we first need to pin down where prices *come from*.

Now we have both sides. Demand tells us how much buyers want to buy at a price. Supply tells us how much sellers want to sell at that same price. Next time we'll put the two together and ask whether there's a price at which those quantities agree.